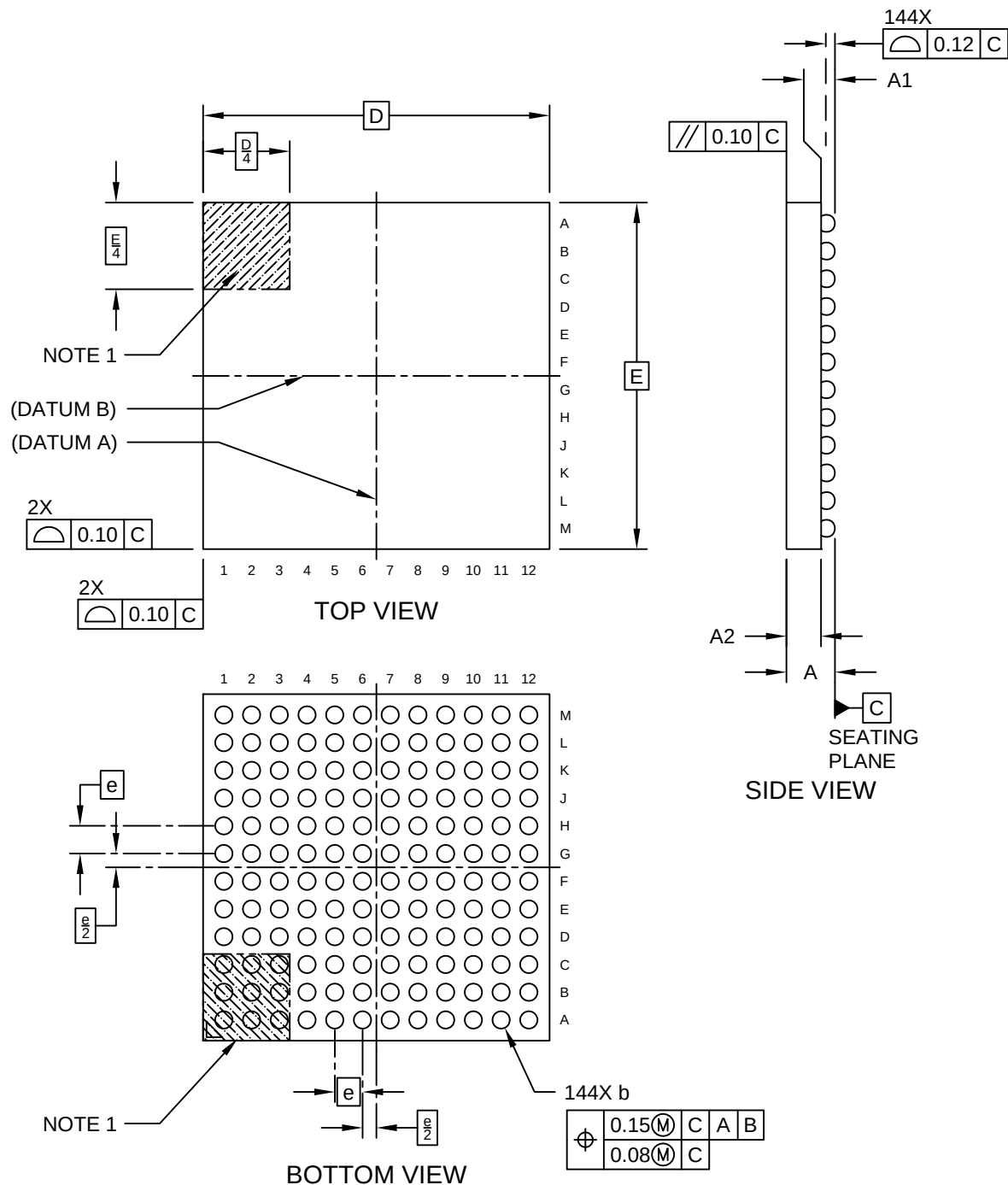


144-Ball Low Profile Fine Pitch Land Grid Array (A8B) - 10X10 mm Body [LFBGA] Atmel Legacy Global Package Code CYL

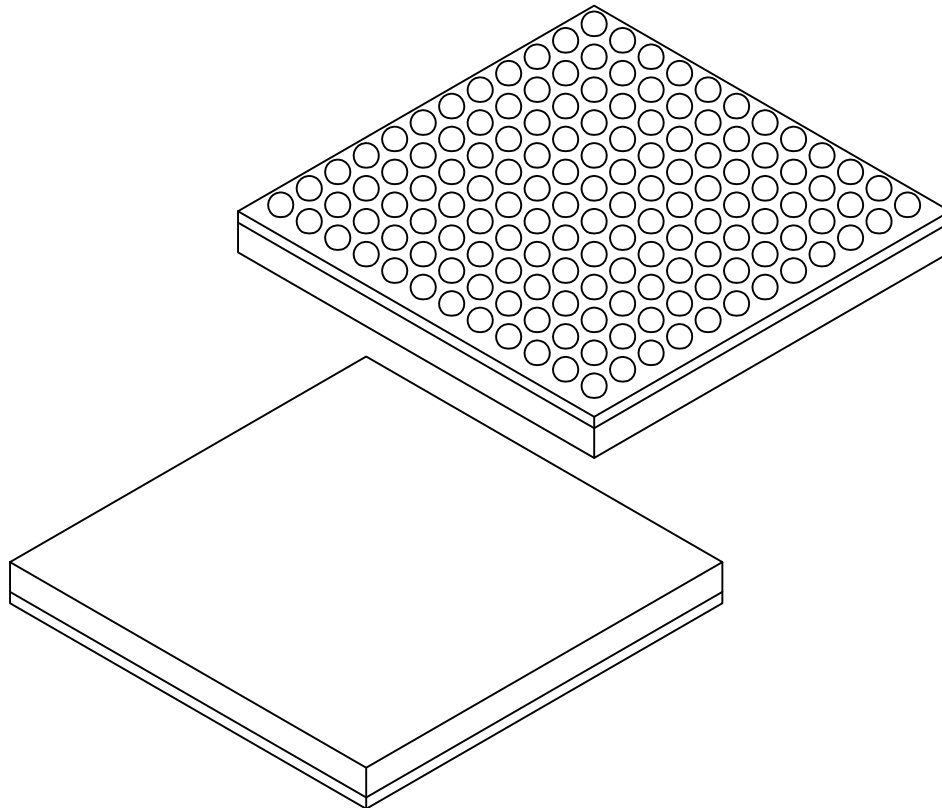
Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Microchip Technology Drawing C04-21116-A8B Rev B Sheet 1 of 2

**144-Ball Low Profile Fine Pitch Land Grid Array (A8B) - 10X10 mm Body [LFBGA]
Atmel Legacy Global Package Code CYL**

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



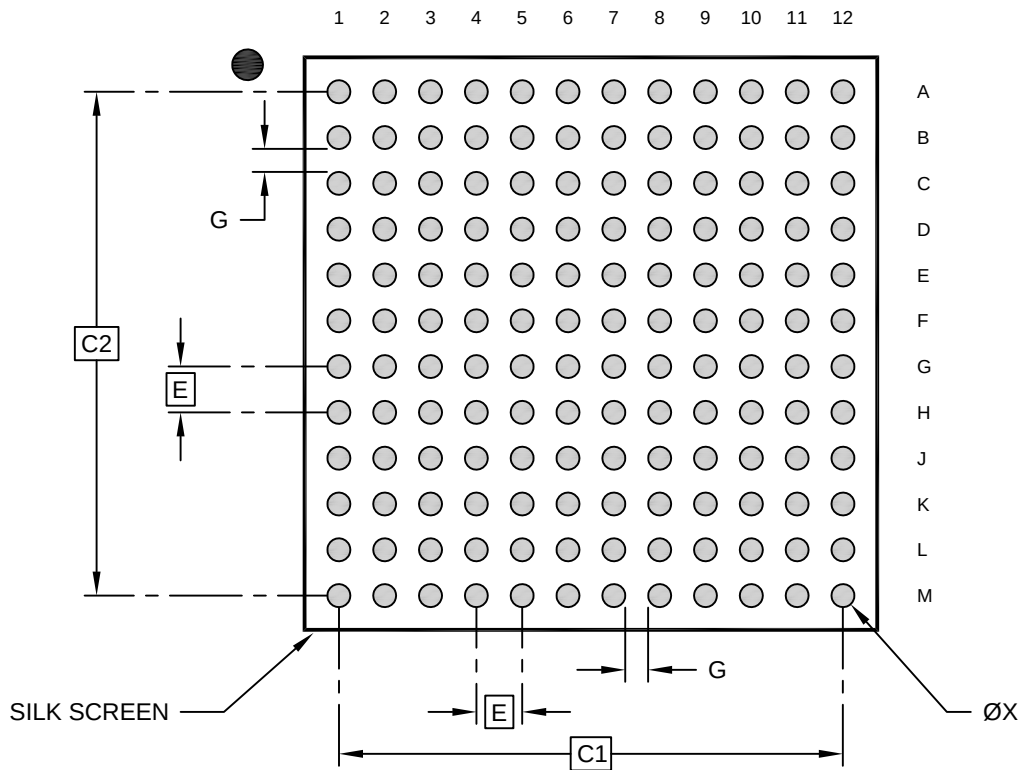
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	144		
Pitch	e	0.80 BSC		
Overall Height	A	-	-	1.40
Ball Height	A1	0.25	-	-
Package Thickness	A2	0.85	-	-
Overall Length	D	10.00 BSC		
Overall Width	E	10.00 BSC		
Terminal Dimeter	b	0.45	0.50	0.55

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

144-Ball Low Profile Fine Pitch Land Grid Array (A8B) - 10x10 mm Body [LFBGA] Atmel Legacy Global Package Code CYL

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

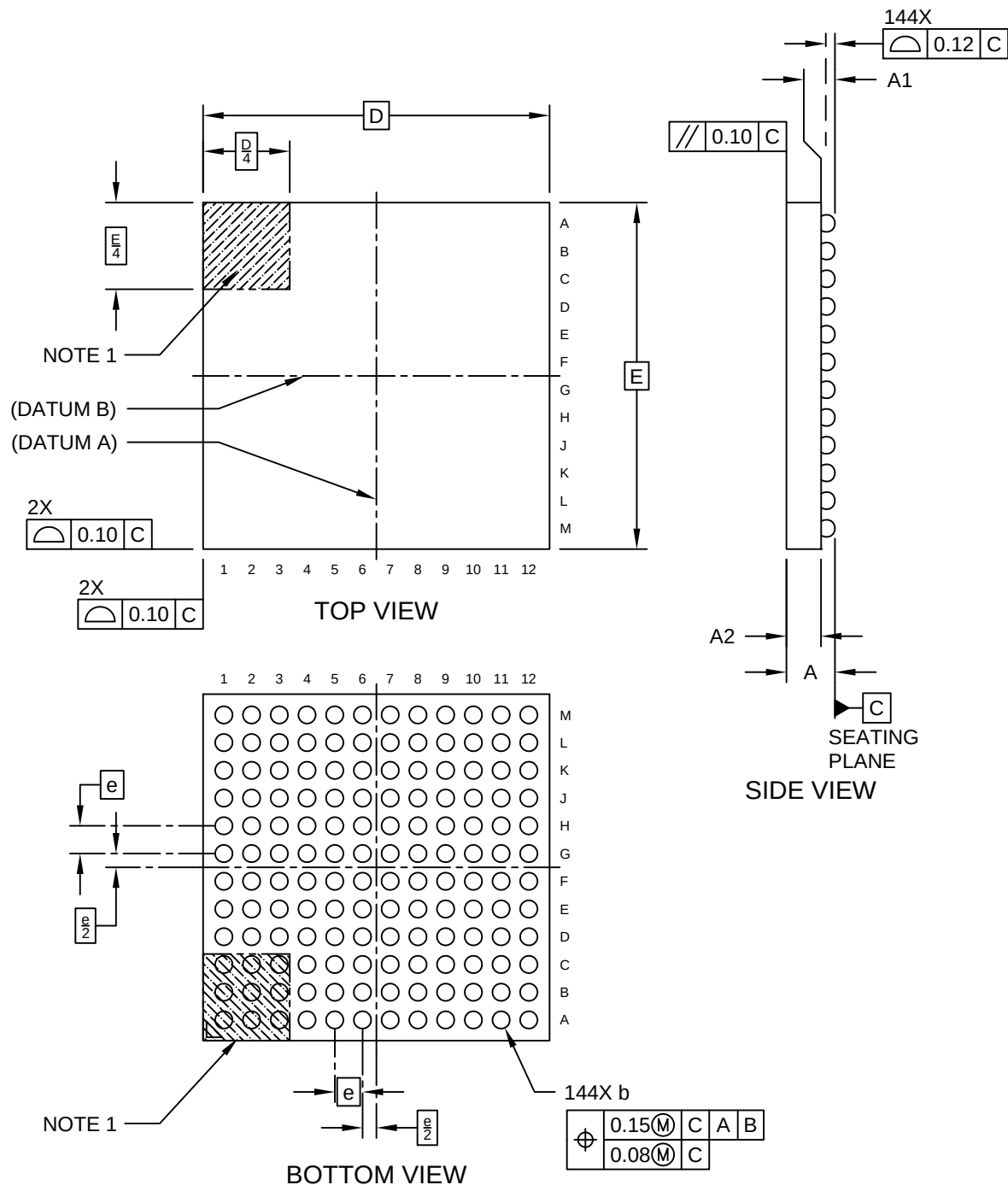
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E		0.80 BSC	
Contact Pad Spacing	C1		8.80	
Contact Pad Spacing	C2		8.80	
Contact Pad Diameter (X144)	X			0.40
Contact Pad to Contact Pad	G	0.40		

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.

144-Ball Low Profile Fine Pitch Land Grid Array (A9B) - 10x10 mm Body [LFBGA] Atmel Legacy Global Package Code CYL

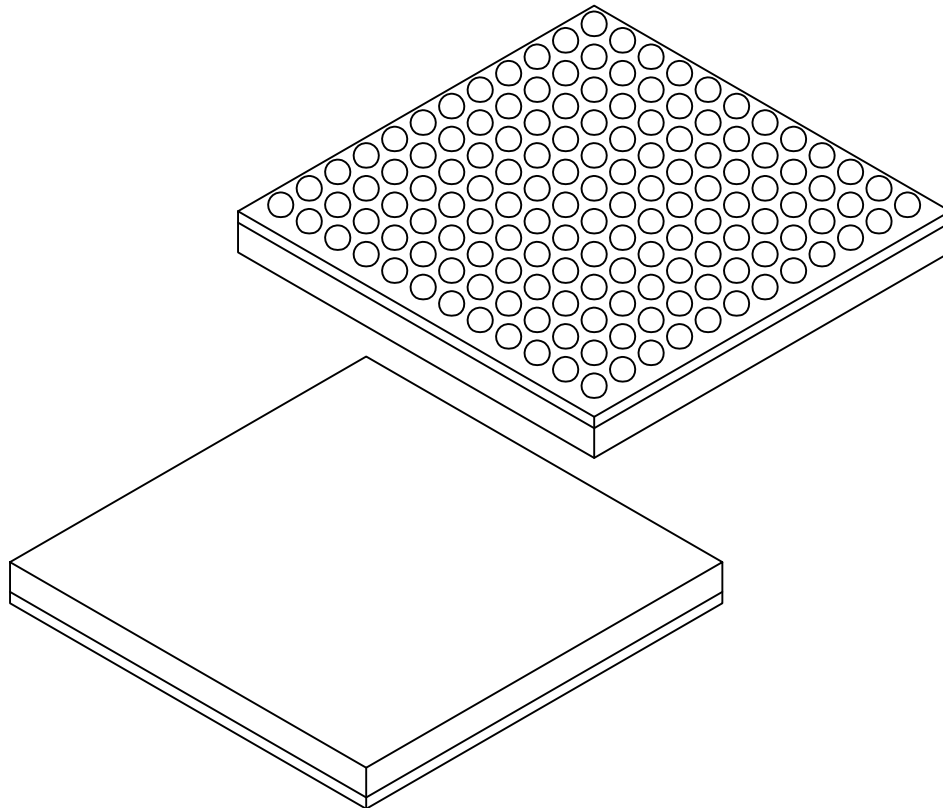
Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Microchip Technology Drawing C04-21116-A9B Rev B Sheet 1 of 2

**144-Ball Low Profile Fine Pitch Land Grid Array (A9B) - 10x10 mm Body [LFBGA]
Atmel Legacy Global Package Code CYL**

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



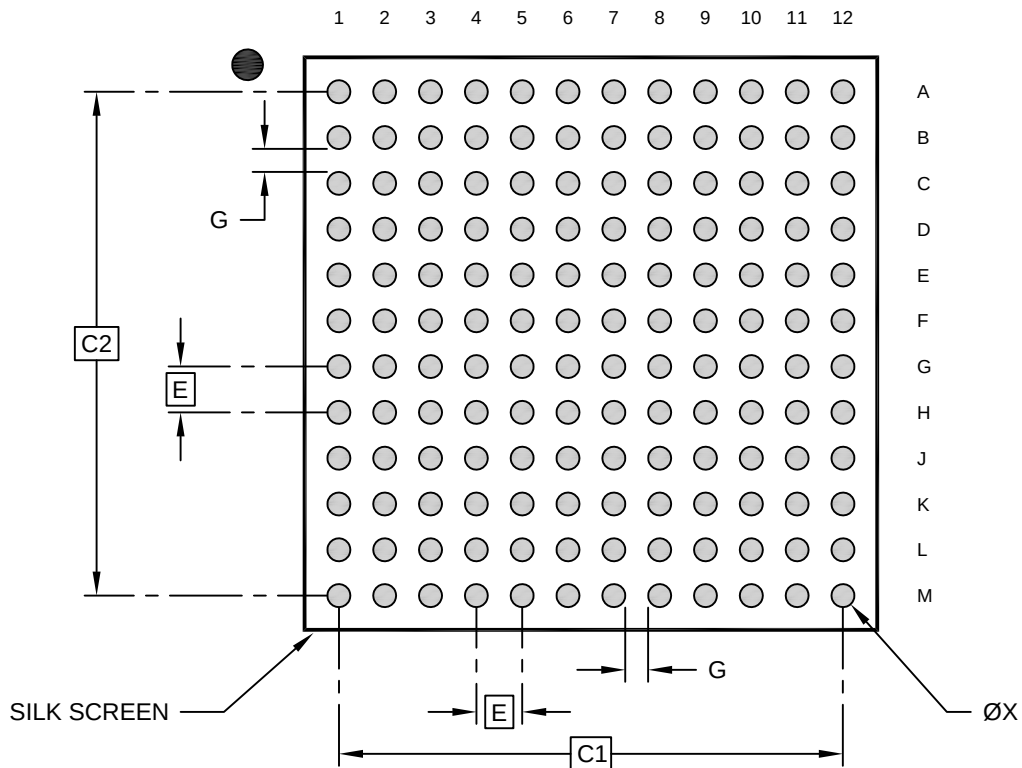
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	144		
Pitch	e	0.80 BSC		
Overall Height	A	-	-	1.40
Ball Height	A1	0.25	-	-
Package Thickness	A2	0.85	-	-
Overall Length	D	10.00 BSC		
Overall Width	E	10.00 BSC		
Terminal Diameter	b	0.45	0.50	0.55

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

144-Ball Low Profile Fine Pitch Land Grid Array (A9B) - 10x10 mm Body [LFBGA] Atmel Legacy Global Package Code CYL

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E		0.80 BSC	
Contact Pad Spacing	C1		8.80	
Contact Pad Spacing	C2		8.80	
Contact Pad Diameter (X144)	X			0.40
Contact Pad to Contact Pad	G	0.40		

Notes:

- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.